

SSD_Uboot Upgrade Guide

1 Overview

SSD20X does not support empty chip upgrade, but provides USB/SD card upgrade function in the Uboot stage. The upgrade can specify the partition that needs to be upgraded, as long as the corresponding upgrade package is prepared and executed according to the following documents.

2. SD card upgrade

Uboot supports SD card by default, just put the corresponding upgrade package (default: SigmasterUpgradeSD.bin) into the root directory of the SD card, and perform the corresponding operation in uboot.

2.1. Build the SD card upgrade package

After the image has been packaged under the project, execute the `./make_sd_upgrade_sigmaster.sh` script and follow the prompts to select the corresponding partition to be upgraded to package and generate the SD card upgrade package.

```
xinhua.ye@szbc6405:~/alkaid_mp/project$ ./make_sd_upgrade_sigmaster.sh
Upgrade [[cis.es? (y/N)y
Upgrade [[set_partition.es? (y/N)y
Upgrade [[ipl.es? (y/N)y
Upgrade [[ipl_cust.es? (y/N)y
Upgrade [[uboot.es? (y/N)y
Upgrade [[logo.es? (y/N)y
Upgrade [[kernel.es? (y/N)y
Upgrade [[rootfs.es? (y/N)y
Upgrade [[miservice.es? (y/N)y
Upgrade [[customer.es? (y/N)y
Upgrade [[appconfigs.es? (y/N)y
UpgradeImage Generating.....
1+0 records in
1+0 records out
16 bytes copied, 0.000117988 s, 136 kB/s
xinhua.ye@szbc6405:~/alkaid_mp/project$
```

After the package is complete, SigmastarUpgradeSD.bin will be generated in the project/image/output/images directory, as shown below. Copy the file to the root directory of the SD card.

名称	修改日期	类型	大小
boot	2021/11/22 18:06	文件夹	
scripts	2021/11/22 18:06	文件夹	
scripts_bin	2021/11/22 18:06	文件夹	
appconfigs.ubifs	2021/11/22 18:06	UBIFS 文件	1,612 KB
auto_update.txt	2021/11/22 18:06	文本文档	1 KB
auto_update_bin.txt	2021/11/22 18:06	文本文档	1 KB
cis.bin	2021/11/22 18:06	BIN 文件	23 KB
customer.ubifs	2021/11/22 18:06	UBIFS 文件	54,932 KB
customer.ubifs.0	2021/11/22 18:06	0 文件	35,840 KB
customer.ubifs.1	2021/11/22 18:06	1 文件	19,092 KB
dummy_pad	2021/11/22 18:06	文件	10 KB
ipl_cust_s.bin	2021/11/22 18:06	BIN 文件	21 KB
ipl_s.bin	2021/11/22 18:06	BIN 文件	22 KB
kernel	2021/11/22 18:06	文件	2,132 KB
logo	2021/11/22 18:06	文件	62 KB
miservice.ubifs	2021/11/22 18:06	UBIFS 文件	7,316 KB
rootfs.ubifs	2021/11/22 18:06	UBIFS 文件	4,464 KB
SigmastarUpgradeSD.bin	2021/11/22 18:06	BIN 文件	70,841 KB
uboot_s.bin	2021/11/22 18:06	BIN 文件	231 KB
upgrade_script.txt	2021/11/22 18:06	文本文档	16 KB

2.2. SD card upgrade operation

SSD20X does not support empty chip upgrade. You can only press enter to enter the uboot console. Enter sdstar in the console to enter the upgrade. After the upgrade is successful, it will automatically restart.

```
SigmaStar #sdstar
```

In addition, you can also set the following env to achieve the effect of automatically detecting and upgrading at startup (that is, you do not need to manually knock sdstar)

```
setenv sdautoupgrade 1;saveenv;
```

3. U disk upgrade

Uboot打开USB驱动支持，然后将对应制作的升级包(默认是：SigmastarUpgrade.bin)放到U盘根目录，在uboot执行对应操作即可。

3.1. 配置Uboot config

使用对应的defconfig配置完后(公版nand配置：make infinity2m_spinand_defconfig)，通过make menuconfig开启如下配置。

Device Drivers下开启如下配置：

```
.config - U-Boot 2015.01 Configuration
Device Drivers

Device Drivers
Arrow keys navigate the menu.  <Enter> selects submenus ---> (or empty submenu)
Pressing <Y> includes, <N> excludes, <M> modularizes features.  Press <Esc> to exit
Legend: [*] built-in  [ ] excluded  <M> module  < > module capable

[ ] Enable Driver Model
    NAND Device Support --->
[ ] Enable watchdog
[*] Support for Host-side USB
    *** USB Host Controller Drivers ***
[ ] xHCI HCD (USB 3.0) support (NEW)
[ ] EHCI HCD (USB 2.0) support (NEW)
[*] USB Mass Storage support
    USB Gadget Support --->
    DFU support ----

↓(+)

<Select>  < Exit >  < Help >
```

MStar drivers下开启如下配置：

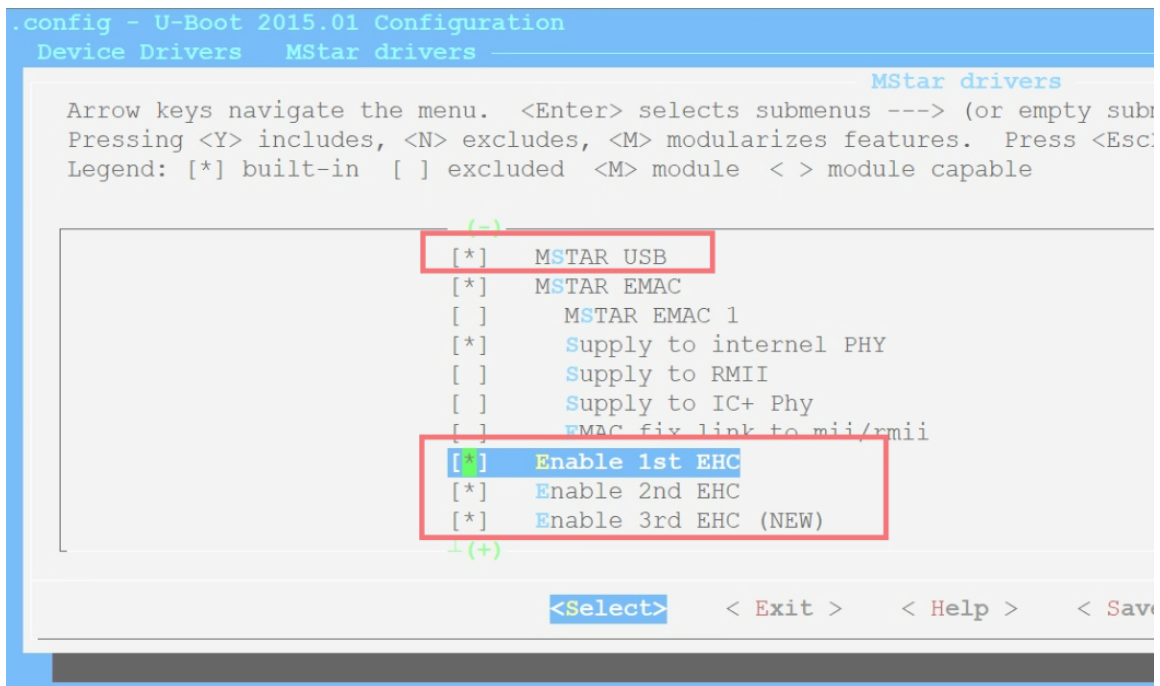
```
.config - U-Boot 2015.01 Configuration
Device Drivers  MStar drivers

MStar drivers
Arrow keys navigate the menu.  <Enter> selects submenus ---> (or empty submenu)
Pressing <Y> includes, <N> excludes, <M> modularizes features.  Press <Esc> to exit
Legend: [*] built-in  [ ] excluded  <M> module  < > module capable

--- MStar drivers
[*] MSTAR ISP NOR FLASH
[*] Save environment to ISP NOR FLASH
[ ] MTD device for ISP NOR FLASH
[*] Enable Mstar partition support
[*] MSTAR SDMMC
[ ] MSTAR eMMC
[*] MSTAR Gpio
[ ] MSTAR NAND
[ ] EYE

↓(+)

<Select>  < Exit >  < Help >  < Save >
```



3.2. 构建U盘升级包

在project下已经将image打包好之后，执行 `./make_usb_upgrade_sigmaster.sh` 脚本按照提示选择需要升级的对应分区打包生成U盘升级包。

```
xinhua.ye@szbc6405:~/alkaid_mp/project$ ./make_usb_upgrade_sigmaster.sh
Upgrade [[cis.es? (y/N)y
Upgrade [[set_partition.es? (y/N)y
Upgrade [[ipl.es? (y/N)y
Upgrade [[ipl_cust.es? (y/N)y
Upgrade [[uboot.es? (y/N)y
Upgrade [[logo.es? (y/N)y
Upgrade [[kernel.es? (y/N)y
Upgrade [[rootfs.es? (y/N)y
Upgrade [[miservice.es? (y/N)y
Upgrade [[customer.es? (y/N)y
Upgrade [[appconfigs.es? (y/N)y
UpgradeImage Generating.....
1+0 records in
1+0 records out
16 bytes copied, 0.000254338 s, 62.9 kB/s
```

打包完成后将在project/image/output/images目录下生成 SigmasterUpgrade.bin，如下图。将该文件拷贝到U盘根目录即可。

名称	修改日期	类型	大小
boot	2021/11/22 18:06	文件夹	
scripts	2021/11/23 10:42	文件夹	
scripts_bin	2021/11/22 18:06	文件夹	
appconfigs.ubifs	2021/11/22 18:06	UBIFS 文件	1,612 KB
auto_update.txt	2021/11/23 10:42	文本文档	1 KB
auto_update_bin.txt	2021/11/22 18:06	文本文档	1 KB
cis.bin	2021/11/22 18:06	BIN 文件	23 KB
customer.ubifs	2021/11/22 18:06	UBIFS 文件	54,932 KB
customer.ubifs.0	2021/11/22 18:06	0 文件	35,840 KB
customer.ubifs.1	2021/11/22 18:06	1 文件	19,092 KB
dummy_pad	2021/11/23 10:42	文件	10 KB
ipl_cust_s.bin	2021/11/22 18:06	BIN 文件	21 KB
ipl_s.bin	2021/11/22 18:06	BIN 文件	22 KB
kernel	2021/11/22 18:06	文件	2,132 KB
logo	2021/11/22 18:06	文件	62 KB
miservice.ubifs	2021/11/22 18:06	UBIFS 文件	7,316 KB
rootfs.ubifs	2021/11/22 18:06	UBIFS 文件	4,464 KB
SigmastarUpgrade.bin	2021/11/23 10:42	BIN 文件	70,841 KB
SigmastarUpgradeSD.bin	2021/11/22 18:06	BIN 文件	70,841 KB
uboot_s.bin	2021/11/22 18:06	BIN 文件	231 KB
upgrade_script.txt	2021/11/23 10:42	文本文档	16 KB

3.3. U盘升级操作

SSD20X不支持U盘空片升级，开机后按enter键进入Uboot控制台。

设置usb升级port口：

如果不确定USB port口，也可以使用usb cmd去检测，如下图检测到storage设备即可：输入 `usb start 0/1/2` 看哪个口可以识别到存储设备（一般0比较常见）

```
SigmaStar # usb start 1
(Re)start USB 1...
Check USB port[1]:
[USB] usb_lowlevel_init++
[USB] USB EHCI LIB VER: 2018.01.12
[USB] Port 1 is Enabled
[USB] TV_usb_init (UTMI Init) ++
[USB] UTMI Base 1f284a00
[USB] UHC Base 1f285000
[USB] USBC Base 1f284e00
[USB] BC Base 1f284c00
[USB] config miu select [70] [f8] [ef] ][ef]
[USB] enable miu lower bound address subtraction
[USB] init squelch level 0x2
[USB] TV_usb_init--
[USB] Usb_host_Init++
[USB] Async base addr: 0x3eb2580
[USB] Reg 0x28: 0x2580 0x3eb
[USB] disable run
[USB] Host Speed:2
[USB] enable aynch
[USB] Usb_host_Init--
[USB] usb_lowlevel_init--
scanning bus for devices...
[USB] interface[0] conf:1 value 8:
1 USB Device(s) found
    scanning usb for storage devices... [USB] no_of_ep: 2
[USB] find bulk ep: 0
[USB] find bulk ep2: 1

[USB] bulk max packet size: ep(in) 0x200, ep2(out) 0x200
[USB] bulk0 is in
max USB Storage Device reached: 1 stopping
1 Storage Device(s) found
SigmaStar #
```

识别到之后设置对应的port口即可，如上截图setenv UpgradePort 1 ; saveenv ;
然后usbstart即进入升级，升级成功后自动重启

```
SigmaStar # setenv UpgradePort 1;saveenv
Saving Environment to NAND...
Erasing NAND...
Erasing at 0x440000 -- 100% complete.
Writing to NAND... OK
SigmaStar # usbstar
UpgradeImage env is null,use default SigmastarUpgrade.bin
Saving Environment to NAND...
Erasing redundant NAND...
Erasing at 0x480000 -- 100% complete.
Writing to redundant NAND... OK
(Re)start USB 1...
Check USB port[1]:
[USB] usb_lowlevel_init++
[USB] USB EHCI LIB VER: 2018.01.12
[USB] Port 1 is Enabled
[USB] TV_usb_init (UTMI Init) ++
[USB] UTMI Base 1f284a00
[USB] UHC Base 1f285000
[USB] USBC Base 1f284e00
[USB] BC Base 1f284c00
[USB] config miu select [70] [f8] [ef] ][ef]
[USB] enable miu lower bound address subtraction
[USB] init squelch level 0x2
[USB] TV_usb_init--
[USB] Usb_host_Init++
[USB] Async base addr: 0x3eb2580
[USB] Reg 0x28: 0x2580 0x3eb
[USB] disable run
[USB] Host Speed:2
[USB] enable aynch
[USB] Usb_host_Init--
[USB] usb_lowlevel_init--
scanning bus for devices...
[USB] interface[0] conf:1 value 8:
1 USB Device(s) found
    scanning usb for storage devices... [USB] no_of_ep: 2
[USB] find bulk ep: 0
[USB] find bulk ep2: 1

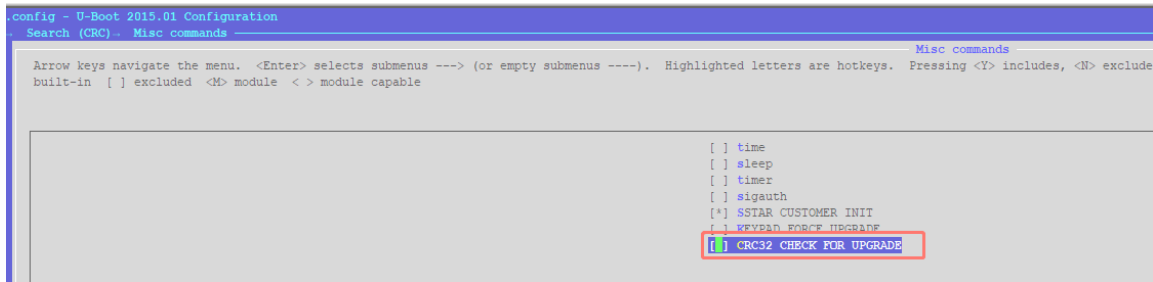
[USB] bulk max packet size: ep(in) 0x200, ep2(out) 0x200
[USB] bulk0 is in
max USB Storage Device reached: 1 stopping
1 Storage Device(s) found
[do_usbstar] Check disable_version_chk !
```

另，还可以通过设置以下env，达到开机自动检测升级的效果（即不需要手动敲usbstar）

```
setenv usbautoupgrade 1;saveenv
```


4. 升级包打开CRC校验

- a. CRC校验针对的是每一个分区数据，升级过程中某个分区数据CRC校验失败，整个升级流程即会被打断
- b. CRC校验需要打开以下config: CONFIG_UPGRADE_CRC_CHECK（默认关闭）



5. 支持版本控制

The current version is just a simple time record of the package:

upgrade_bin_version=11261056

- a. When upgrading, the version recorded in the upgrade_bin_version on the env on the board will be compared, and only when the version is different will it be upgraded (the size will not be compared, and it will be upgraded if it is not the same)
- b. You can turn on version verification by setting env: setenv enable_version_chk 1; saveenv;

6. USB/SD card upgrade support UI progress bar display

The following config needs to be opened: CONFIG_SSTAR_UPGRADE_UI (off by default)

Arrow keys navigate the menu. <Enter> selects submenus ---> (or empty submenus ----). Highlighted letters are hotkeys. Pressing <Y> includes, <N> excludes built-in [] excluded <M> module <> module capable

```
--- MStar drivers
[ ] SSTAR IR
[*] MSTAR ISP NOR FLASH
[ ] Save environment to ISP NOR FLASH
[ ] MTD device for ISP NOR FLASH
[*] Enable Mstar partition support
[*] MSTAR SDMMC
[ ] MSTAR eMMC
[*] MSTAR Gpio
[ ] MSTAR NAND
[ ] EYE
[*] MSTAR USB
[*] MSTAR EMAC
[ ] MSTAR EMAC 1
[*] Supply to internal PHY
[ ] Supply to RMII
[ ] Supply to IC+ Phy
[ ] EMAC fix link to mii/rmii
[*] Enable 1st EHC
[*] Enable 2nd EHC
[*] Enable 3rd EHC
[ ] Enable USB lan module
[*] Enable AESDMA
[*] MSTAR SPINAND
    Read mode (Use Quad read) --->
    Write mode (Use single write) --->
[*] Save environment to NAND FLASH
(_spinand) Append postfix name to compressed file
[*] SSTAR DISP
[ ] SSTAR DISP HDMIIX VGA
[*] SSTAR PNL
[ ] SSTAR HDMIIX
[*] SSTAR JPD
[*] SSTAR RGN
[*] SSTAR MSPI
[*] SSTAR SAR
[*] SSTAR WDT
[*] SSTAR EWM
[ ] SSTAR UPGRADE UI
```